

CLASS 01 / AUG 25

ARC 3133

VISUAL IDENTITY

HOW THE WORK WILL LOOK, BEFORE THERE IS ANY WORK

NOTHING DUE

GSM ISSUED TODAY

WHAT THIS COURSE IS

Computational methods for architectural representation and digital fabrication.

WHAT YOU WILL DO

Set your own graphic standards.
Describe geometry with parameters.
Repeat it, and vary it.
Work with sampled and volumetric data.
Turn all of it into drawings and physical parts.

WHAT CARRIES THROUGH

Visual quality stays central all semester — composition, hierarchy, line weight, colour, rendering, photography.

A working graph supports those decisions. It does not replace them.

EACH PROJECT CARRIES ITS GEOMETRY INTO THE NEXT. YOU NEVER START FROM AN UNRELATED OBJECT.

THREE PARADIGMS

Each one a strict extension of the one before it.

01 PROCEDURAL

Form as a fixed sequence of operations.
You define the steps.

Output is expected.

02 PARAMETRIC

Relationships between inputs and form.
You define the rules of variation.

Output is a family — you know the range, not each member.

03 GENERATIVE

A rule applied to its own output, repeatedly. You define the rule and the starting state.

Output is unknown, within a spectrum you set.

A procedure is a fixed recipe. A parametric system is a recipe with dials. A generative system reads its own output and decides what to do next.

FOUR PROJECTS

Ten short assignments, assembled into four. Nothing is graded twice.

P1

Geometry — from operations to construction

CSG massing sequence → Panel from scratch → First print

21%

P2

Repetition and response — arrays to attractors

Arrays and conditional control → Attractor field study

14%

P3

Spatial arrays and assembly

3D array and connection study → Modular print assembly

14%

P4

Samples, fields and sections

Point-cloud analysis → Volume, section and light → Sectional fabrication

21%

You do the short assignment, then revise it inside its project. The project itself adds no second grade — the revision is what the booklet rewards.

HOW A CLASS RUNS

The same four things, every week.

01

REVIEW OF WORK

We open the files. Bring the editable file, not a screenshot of it.

02

DEMONSTRATION

A short tutorial or discussion of the week's unit.

03

GUIDED PRACTICE

You build the small exercise while help is in the room.

04

WORKING CRITIQUE

We look at what is on screen and say what it needs.

YOU WILL BE ASKED TO CHANGE ONE PARAMETER AND PREDICT WHAT HAPPENS. FROM THE FIRST REVIEW.

WHY THIS COMES FIRST

Every sheet you make for the next fifteen weeks uses the system you build now.

A DRAWING IS READ BEFORE IT IS UNDERSTOOD

Hierarchy is what makes an architectural drawing legible — not detail, and not effort.

Line weight, spacing and restraint do more work than anything you will model this semester. A good system makes weak work readable. A bad one hides good work.

WHAT A SYSTEM IS

Page size and margins.

A column grid.

A type hierarchy — three levels, no more.

A line-weight set — four weights.

A palette of three colours, and the black is one of them.

Three typefaces is the ceiling. Two is better.

DECIDE IT ONCE, APPLY IT EVERY WEEK, REVISE IT AT THE MIDTERM. IT IS NOT REMADE PER SHEET.

THREE MANUALS WORTH STEALING FROM

This document type has a history. Each one is a page grid, a type hierarchy, and a rule for every case that comes up.

Add: a spread from the NYCTA manual - a signage case with its dimensioned rule.

NYCTA GRAPHICS STANDARDS

Massimo Vignelli and Bob Noorda, Unimark - 1970 - archive.org/details/nycta-gs-manual

A binder of cases. Every sign in the New York subway, drawn at size, with the rule beside it. The manual is the design - the signs are what falls out of it.

Add: a page from the NASA manual - the mark with its clearance and placement rules.

NASA GRAPHICS STANDARDS

Richard Danne and Bruce Blackburn - 1975 - archive.org/details/NASA_Graphics_Standards_Manual

One mark, and a hundred pages of where it is allowed to go. What is being designed is the consistency, not the logotype.

Add: the Munich 1972 pictogram sheet, and the construction grid behind one figure.

MUNICH 1972

Otl Aicher - 1967 to 1972 - otlaicher.de, 'the rainbow games'

A grid, a fixed set of angles, and one component varied across a whole field of pictograms. You will build a system with this exact structure in Week 4.

YOURS IS FOUR PAGES, NOT THREE HUNDRED. THE PRINCIPLE IS IDENTICAL: DECIDE ONCE, WRITE IT DOWN, APPLY IT EVERY TIME.

All three are online in full, linked on the U01 lesson page and under Reading. Look at how much of each one is rules rather than pictures.

A VISUAL IDENTITY IS A PARAMETER SET

A short list of values, decided once. Every sheet this semester reads them.

```
PAGES      = 4                # the whole manual
SHEET      = (17, 11)         # inches
MARGIN     = 0.75
COLUMNS   = 6

TYPEFACES  = 2                # 3 is the ceiling
TYPE       = {"title": 24, "heading": 14, "body": 9}
WEIGHTS    = [0.13, 0.35, 0.70] # thin, medium, heavy
COLORS     = 3                # the black counts as one
PALETTE    = ["#000000", "#FFFD7", "#C6F035"]
```

```
# every sheet this semester reads these values
```

WHAT IT MAPS ONTO

composition
grid
hierarchy
typography
line weight
palette

A BINDER OF RULES AND THIS SHORT LIST ARE THE SAME DOCUMENT, AT DIFFERENT SCALES.

ISSUED TODAY

GSM — GRAPHIC STANDARDS MANUAL

DUE WEEK 8 — Oct 13 / 7%

Build the graphic system you will communicate every later project with. It is the only assignment in the course that does not require a computational graph — it is judged purely on whether the system works and whether you used it.

DECIDE

THE SYSTEM

Grid, margins, type, line weights, colour.
Written down, not improvised.

TEST

ON REAL WORK

A diagram and a rendered image, made with
the rules you just set.

REVISE

ALL SEMESTER

Due at the midterm, revised again for the
booklet.

REQUIREMENTS — GSM

Assessed on visual quality and hierarchy 40%, consistency and usability 30%, tested applications 20%, completeness 10%.

01

Four pages, and no more. The manual is the graphic system itself, shown in use: rules with a tested application beside each one, not a description of intentions.

02

One page per decision: (1) the template — sheet size, margins, grid and columns, with the placement rules; (2) typography and hierarchy; (3) color and line weights; (4) annotation, captions and image treatment, demonstrated on a real sheet from this course.

03

Use no more than three typefaces, three colors in total — black, the paper and every accent count toward the three — four hierarchy levels and four line weights.

04

Study the three manuals shown in Class 01 — NYCTA, NASA and Munich 1972 — and name one rule you took from them. All three are linked in full under reading and documentation.

05

Apply the manual to every sheet; revise it in response to actual use and critique.

CHECKPOINT · WEEK 2 — Sep 1 — Bring a sample sheet for an ungraded identity check.

NOW — YOUR FIRST SHEET

Rest of the session: set the page up and put something on it.



Page size set to 17 × 11 inches, with margins and a column grid



Type hierarchy chosen — three levels, and a reason for each



A line-weight set, from hairline to heaviest — four weights



Three colours you can defend, black included, that survive being printed



One sample sheet, with one decision about hierarchy you can explain

LEAVE WITH A SHEET. IT DOES NOT HAVE TO BE RIGHT YET.

BEFORE NEXT CLASS

WATCH

U02 — Transformations and CSG. The node editor, parameters, and the three Boolean operations. Check the unit page; watch a recording only once a link is posted.

INSTALL

Blender, and set up a project folder you can open from a clean machine. File organization is part of the workflow, not an afterthought.

BRING

Your sample sheet and the file it came from.

LOOK

Three architectural drawings you think are well made. Be ready to say what makes each one readable.